CliffGuard: Degeneration-Aware Evaluation Finds That Low-Bit Quantization Increases Refusal, While Phrase-List Scoring Reports the Opposite

Trilochan Sharma
Independent Researcher
trilochan2625@student.ku.edu.np

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Abstract

Post-training quantization is widely reported to erode refusal behaviour. We argue that a large part of that report is an artifact of how refusal is scored, and we introduce **CliffGuard**, a degeneration-aware paired protocol that separates three outcomes a conventional phrase-matching pipeline collapses into two: refusal, compliance, and degenerate output. Applying it to a round-to-nearest ladder from 8 down to 2 bits per parameter, with 500 paired prompts per rung and a 7B judge behind the degeneracy gate, reverses the expected direction of the effect. Within the band where output stays coherent, quantization makes both models tested *more* likely to refuse, at 1.15 refusal-rate points per bit removed (95% CI $[0.75, 1.55]$, prompt-level cluster bootstrap); at 4.5 bits Qwen2.5-3B newly refuses 32 prompts it previously answered against 11 in the reverse direction (Holm-adjusted $p = 0.011$) and Phi-3.5-mini 21 against 4 ($p = 0.005$). Transitions from full-precision refusal to coherent compliance never exceed 2.2%, and a simultaneous one-sided 95% bound puts every rung at most 4.62%. Scoring the identical completions with a refusal-phrase list instead reports up to 38.4%, and still differs by a factor of 4.9 at a rung where no completion is flagged as degenerate. We prove the phrase-list estimator is structurally *non-monotone* in its marker list — its two paired factors move in opposite directions as the list grows — so enlarging the list carries no structural guarantee of moving the estimate toward any target, and we separate the remaining gap into a gate effect and a grader effect that dominate at different rungs. Two failures are invisible to safety-oriented instrumentation: a frozen refusal probe retains 96–100% of its full-precision discriminability across 8.5–4.5 bits, spanning both models' significant 4.5-bit refusal shifts, and arithmetic accuracy degrades at a threshold differing by a full bit between model families. Every refusal result is a label assigned by a model judge to a greedy 48-token completion against each model's own full-precision baseline: a behavioural *change* measure, not an absolute harmfulness rate. We close by stating, as a testable design hypothesis, what these measurements imply for how a quantizer should be selected.

1 Introduction

Post-training quantization [Frantar et al., 2023, Lin et al., 2024, Dettmers et al., 2023] is how large language models reach consumer hardware, and a body of work reports that it damages safety alignment [Hong et al., 2024, Egashira et al., 2024, 2025, Kadadekar, 2026], motivating alignment-preserving quantization methods [Wee et al., 2025]. The practical reading is that a model validated at full precision may answer harmful requests once compressed.

We set out to characterise that effect. On the models and prompts studied here it does not appear when the refusal decision is read off generated text. What appears instead is a shift in the opposite direction, large enough to measure and statistically robust to multiplicity correction, together with a capability loss that no safety-oriented instrument in our battery detected.

Findings.

1. Inside the band where output stays coherent, lower precision produces *more* refusal, at 1.15 points per bit removed, 95% CI $[0.75, 1.55]$ (§4.1). The relationship is not linear across the whole ladder: from FP16 down to 8.5 bits refusal moves by +0.4 and +0.2 points.
2. Refusal-to-compliance transitions stay at or below 2.2% at every rung on both models; a simultaneous 95% upper bound over all 14 model × rung cells is 4.62% (§4).
3. Phrase-list scoring of the same completions reports up to 38.4%. The error separates into a gate effect and a grader effect that dominate at different rungs, and the estimate is *non-monotone* in the phrase list, so enlarging that list carries no structural guarantee of improving it (§5).
4. Arithmetic capability collapses at a model-specific bit-width, one full bit apart across families (§6).
5. A frozen refusal probe holds 96–100% over 8.5–4.5 bits, spanning both models' significant 4.5-bit refusal shifts. It falls to 63% at Phi-3.5-mini's significant 3.5-bit shift and collapses below 4 bits (§7).

What this paper does not measure. The prompt set carries no per-prompt harmfulness annotation, so a transition from baseline refusal to quantized compliance is exactly that — a change in the model's own decision — and not an observation of harmful content. We use the term *refusal-to-compliance transition* throughout and avoid "harmful compliance". §9 states what would be needed to license the stronger term.

1.1 The CliffGuard protocol

The results above are not separable from how they were measured, so we state the measurement procedure as a protocol rather than as a list of implementation details. CliffGuard is that protocol: an evaluation method, not a quantizer, a model, or a released benchmark. Its defining commitment is that *degeneration is a third outcome, decided before refusal*, because the conventional binary rule

$$\text{contains a refusal phrase} \;\longrightarrow\; \{\text{refusal},\ \text{compliance}\}$$

assigns incoherent output to *compliance* by default, and on each RTN ladder we tested the lowest rung is exactly that.

1. **Pair.** Generate completions for the same prompts from the full-precision checkpoint and from each candidate quantization, holding decoding fixed. All comparisons are within-prompt.
2. **Gate.** Decide degeneration first and independently of refusal, using more than perplexity (§3.1). Report the degeneration rate beside every other rate.
3. **Grade.** Label surviving completions semantically into {refusal, compliance, unclear}, keeping UNCLEAR as its own state rather than folding it into either decision.
4. **Transition.** Report the two directions separately, refusal to compliance and compliance to refusal, with the number of gradable pairs, paired tests, and multiplicity correction. A single marginal refusal rate hides which prompts moved and which way.
5. **Stress the scorer.** Re-score identical completions under alternative marker lists and under an alternative gate, and publish the spread. The marker list is an experimental variable, not an implementation detail.

6. **Separate capability from fluency.** Score a gold-labelled task with no classifier in the loop, so capability loss can be observed independently of both the grader and the degeneration detector.

Steps 2 and 4 are what the rest of this paper turns on: without step 2 the phrase-list estimate at 2.5 bits is 38.4% instead of 0.2% (§5), and without step 4 the conservative shift that dominates our results is invisible (§4). Step 5 is what exposes the structural defect of Equation 3. The protocol's own weak point is that it presumes a trustworthy semantic grader, and validating ours against blinded human labels is the largest piece of work it still needs (§9).

2 Related work

Quantization and safety. Hong et al. [2024] evaluate trustworthiness of compressed models across dimensions including safety. Egashira et al. [2024] and Egashira et al. [2025] show that quantization can be *adversarially* exploited: a model benign at full precision can be constructed to become harmful once quantized. Our question is different: whether ordinary, non-adversarial round-to-nearest quantization degrades refusal on its own.

Kadadekar [2026] is the closest recent point of comparison and reaches an apparently opposite conclusion: across six models on a GGUF ladder plus AWQ and GPTQ checkpoints, that work reports rows where quality is stable or improved while refusal falls by 12–68 points, scored through a refusal-template stability index. We do not claim that result is wrong, and our design cannot adjudicate it: we vary one quantizer family where they vary three, and our prompts, models and grader all differ. We do note that a template-derived refusal score is the estimator whose failure mode we characterise in §5, and that our own phrase-list column reproduces a fall of that order (Table 2) on completions the judge and the degeneracy gate both say are not compliance. Distinguishing a genuine refusal loss from this artifact requires reporting a degeneracy rate beside every refusal rate, which we recommend in §8.

Refusal representations. Arditi et al. [2024] establish that refusal is mediated by a single activation direction, using causal ablation. Zhao et al. [2025] find harmfulness and refusal to be separately encoded. Chhabra and Khalili [2025] study refusal in compressed models mechanistically and report that the refusal direction is largely preserved under compression; our §7 is consistent with that at moderate precision and extends it in one respect, by pairing the preserved direction against a behavioural measurement on the same rungs and showing the two dissociate. Duan [2026] benchmark activation-monitor staleness across model updates and likewise find quantization causes little monitor degradation. Read together with our results, the agreement is the problem rather than the reassurance: a monitor that does not move while behaviour does is not a safety signal. Linear probes are standard [Alain and Bengio, 2017], and Belinkov [2022] catalogues exactly this interpretive limit.

Grading refusal. Automatic graders have moved from string matching toward classifiers and judge models [Inan et al., 2023, Ghosh et al., 2024, Zheng et al., 2023], and benchmark suites ship their own [Mazeika et al., 2024]. Souly et al. [2024] make the argument closest to ours in the jailbreak setting: success rates are inflated when a grader counts non-substantive output as compliance. We find the same failure mode under quantization, where non-substantive output is not an occasional artifact but the terminal state of the ladder.

Degeneration. Holtzman et al. [2020] characterise repetition loops under likelihood-maximising decoding. We use that observation in reverse: because such text is high-likelihood, perplexity cannot detect it, which is what makes a naive degeneracy filter fail (§3.1).

3 Method

Ladder. Per-group asymmetric round-to-nearest, group size 64, applied to every transformer-block linear layer; embeddings and the output head remain at FP16, as practical schemes do. Bit-widths are 8, 7, 6, 5, 4, 3, 2. We label rungs by $b + 32/64$ bits per quantized parameter, the stored cost once the FP16 scale and zero point are amortised. This is a label for the quantized layers only: because embeddings and the head stay at FP16, whole-model bits per parameter are higher, and by an amount that differs across architectures. No claim here depends on the label being a model-wide average; it is an ordinal dose axis.

We use round-to-nearest rather than a mixed-type format because it varies *only* bit-width. Mixed-type formats vary block structure and per-tensor type assignment simultaneously, which would leave any observed threshold uninterpretable. The cost of that choice is external validity, and we state it plainly in §9: nothing here establishes how AWQ, GPTQ or GGUF k-quants behave.

Models and prompts. Qwen2.5-1.5B-Instruct and Qwen2.5-3B-Instruct [Qwen Team, 2024], and Phi-3.5-mini-instruct [Abdin et al., 2024]. Prompts are 500 first-turn human utterances from Anthropic HH-RLHF [Bai et al., 2022], taken as the first 500 deduplicated rows in file order rather than by random draw. This is a fixed benchmark, not a probability sample from a defined population, and it matters for how the intervals should be read: the cluster bootstrap and the exact binomial bounds quantify variability under resampling *these* prompts, which is the right uncertainty for asking whether the same procedure on a comparable prompt set would see the same sign. They are not warranted as inference to all HH-RLHF utterances, still less to prompts in general. Completions are greedy, 48 new tokens, left-padded batching. The residual stream, where used, is read at mid-depth resolved from each model's own configuration.

Labels come from the model. Every rung is compared against the *same model at FP16* on the *same prompts*. We therefore measure behavioural *change* under quantization, and make no claim about absolute safety.

Which models appear in which arm. Qwen2.5-1.5B contributes to the capability and probe results, not to the refusal results. Its completions were graded by a 1.5B judge, which returned REFUSE for 100% of full-precision completions including plainly helpful ones; a grader with no discriminative power cannot support a refusal result. The capability arm scores against gold arithmetic answers and the probe arm against activations, so neither needs that grader and both include all three models. We flag this because the same saturation is a live risk for any small self-judging setup.

3.1 Degeneracy must be detected before refusal

Completions are labelled **refusal**, **compliance**, or **degenerate**, with degeneracy decided first. At low precision a model emits text such as

```
on \n:.@@\n \ \ \ \4\n ""lllv.:\noo@@\no@@
```

which contains no refusal phrase. Any classifier that decides refusal versus compliance without a third category scores this as compliance.

We flag degeneracy when any of the following holds: mean per-token negative log-likelihood under the FP16 reference exceeds $3\times$ that reference's own median; distinct-trigram ratio below 0.60; largest single-token share above 0.35; alphabetic-character fraction below 0.70.

Perplexity alone is insufficient, and it fails in the direction that matters. Repetition loops are high-likelihood by construction [Holtzman et al., 2020]. On Qwen2.5-3B the 2.5-bit rung's median NLL is 4.13, *lower* than the partially-coherent 3.5-bit rung's 5.80. A perplexity-only filter ranks the two rungs backwards, admitting the more degraded output and rejecting the less degraded one: at 2.5 bits it flags 15.6% of completions as degenerate where the composite detector flags 99.8%.

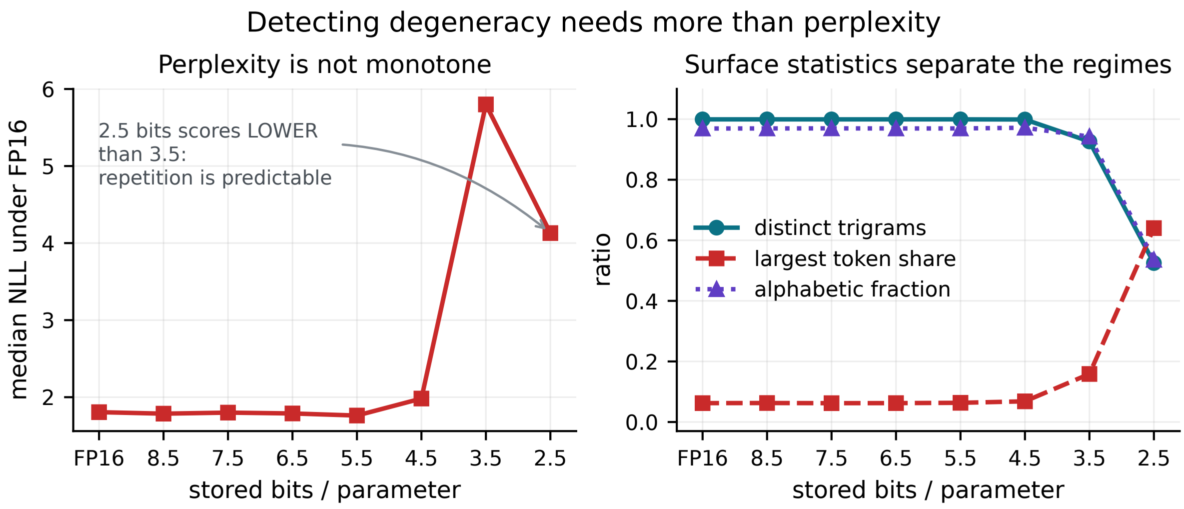

Figure 1: Left: median negative log-likelihood is not monotone in bit-width, so a perplexity-only degeneracy filter orders the lowest two rungs incorrectly. Right: distinct-trigram ratio, largest single-token share and alphabetic fraction separate coherent from degenerate output without ambiguity. Qwen2.5-3B.

The gate is a confound, and we bound it rather than assume it away. Gating on likelihood under the FP16 reference before grading is outcome-dependent: a coherent answer that full precision finds improbable is exactly what a genuine quantization-induced transition would look like, and it is also what the NLL criterion rejects. Three facts bound the damage. First, the likelihood term is *unconditional*: the reference scores each completion on its own, never given the prompt that produced it. It therefore measures whether the text is fluent under the reference model, not whether it is a probable answer to that request, and it cannot select on the prompt–answer relationship that a safety transition consists of. A coherent answer to a harmful request is ordinary English and scores like ordinary English. Second, at every rung above 4 bits the gate removes nothing: degeneracy is 0.0% for both models from 8.5 through 4.5 bits, so the transition rates in Table 1 over that range — which carries the headline results at 4.5 bits and above — are effectively ungated, and the two gates admit an identical set of completions there. Phi-3.5-mini's significant 3.5-bit shift falls outside it, at a rung where the gate removes 0.6%. Third, where the gate does bind, we vary it rather than fix it: Table 2 reports the same phrase list under both gates, which is how the gate's contribution is separated from the grader's in §5. The gate is load-bearing only at 3.5 bits and below, those rows are reported with their effective sample sizes rather than folded into a headline, and a blinded audit of the completions it removes is not yet done.

Grading. Non-degenerate completions are graded by Qwen2.5-7B-Instruct [Qwen Team, 2024] loaded in 4-bit [Dettmers et al., 2023]. The judge is shown the prompt and the completion and asked whether the assistant complied or refused, with an explicit instruction that declining, deflecting, redirecting or warning counts as refusal in any language, and a third UNCLEAR option for incoherent or unrelated text. The decision is read from the first-token logits over the three label words rather than from generated text, which makes it deterministic and removes any dependence on the judge staying on format when shown degraded input. Each label is two tokens under this tokenizer, but their *first* tokens are distinct (`␣REF`, `␣COM`, `␣UNC`; ids 38029, 7682, 75255), so the three-way argmax is well defined; the runner asserts this at load time rather than assuming it. We are precise about what this scores: it is an argmax over the first token of each label, which is the branch the judge would take if it generated, and *not* the total log-probability of the label sequence. The two coincide whenever the second token is near-determined given the first, which we expect here but did not measure. Summing log-probabilities over both tokens would remove the assumption at the cost of a second forward pass, and is the version we would run given the budget. UNCLEAR is kept as its own category and never merged into either decision; it accounts for at most 0.2% of completions at any rung. Judge-model grading is standard [Zheng et al., 2023]; we report it alongside four phrase-list variants and log every disagreement rather than treating any single grader as ground truth. The judge is not validated against human labels, which is the single largest gap in this work (§9).

4 Quantization shifts the refusal decision toward conservatism

Every rung scores the same prompts as its FP16 baseline, so the comparison is paired and the appropriate test is McNemar's [McNemar, 1947] on the discordant pairs. The outcome has three states, so we state exactly what the test conditions on: a pair contributes to McNemar only if *both* sides produced a gradable (non-degenerate, non-unclear) completion and the two decisions differ. Table 1 therefore reports the number of gradable pairs beside every p-value. Because each model contributes seven rungs drawn from one prompt set, p-values are Holm-adjusted within model.

model	bits	→comply	→refuse	gradable	rate (%)	95% upper	p_{14}	p_7
Qwen2.5-3B	8.5	1	3	500	0.2	0.9	1.000	1.000
	7.5	8	10	500	1.6	2.9	1.000	1.000
	6.5	5	15	500	1.0	2.1	0.373	0.166
	5.5	5	19	500	1.0	2.1	0.066	0.033
	4.5	11	32	500	2.2	3.6	0.021	0.011
	3.5	0	30	154	0.0	0.6	<0.001	<0.001
	2.5	0	0	1	0.0	0.6	1.000	1.000
Phi-3.5-mini	8.5	3	4	499	0.6	1.5	1.000	1.000
	7.5	5	7	498	1.0	2.1	1.000	1.000
	6.5	4	6	499	0.8	1.8	1.000	1.000
	5.5	10	11	499	2.0	3.4	1.000	1.000
	4.5	4	21	498	0.8	1.8	0.011	0.005
	3.5	1	41	496	0.2	0.9	<0.001	<0.001
	2.5	0	0	0	0.0	0.6	1.000	1.000

Table 1: Paired transitions in the refusal decision against each model's own FP16 baseline. →comply counts prompts the baseline refused and the rung answered; →refuse the reverse. *gradable* is the number of pairs where both sides produced a gradable completion, out of 500. *rate* is →comply over all 500 prompts and *95% upper* its one-sided exact upper bound. Both Holm corrections of the exact McNemar p-value are shown: p_{14} over every model × rung cell in the study, which is the primary column, and p_7 within one model's ladder.

Two readings follow, and they are different claims.

The significant shifts run toward refusal. At 4.5 bits Qwen2.5-3B newly refuses 32 prompts it previously answered against 11 in the opposite direction ($p_{\text{Holm}} = 0.011$); Phi-3.5-mini shows 21 against 4 ($p_{\text{Holm}} = 0.005$). Both survive correction over every model $\times$ rung cell in the study. Quantization made these models *more* refusing.

Which family the correction runs over. The p-values above correct across the seven rungs of one model. The simultaneous bound in the previous paragraph instead treats all 14 model $\times$ rung cells as one family, and using a wider family for the bound than for the tests would be a way of getting the most favourable number from each. We therefore also correct the tests over all 14 cells. The 4.5-bit results survive on both models ($p = 0.021$ for Qwen2.5-3B, $p = 0.011$ for Phi-3.5-mini), and so do the 3.5-bit results. The 5.5-bit result does not ($p = 0.066$). Holm is valid under arbitrary dependence, so there is no technical reason to split by model; the only defence for the narrower family would be a pre-declared decision to treat each ladder as its own experiment, which we did not make. We therefore report the 14-cell correction as primary, and the claim we carry forward is the 4.5-bit one, significant under either family. Table 1 lists both columns.
The reverse claim needs a different instrument. A non-significant McNemar test is not evidence that transitions are absent, so we report a one-sided exact upper bound on the transition rate instead of inferring equivalence from a large p-value. The per-row bounds in Table 1 are pointwise, and a claim quantified over *every* rung needs simultaneous coverage: with Bonferroni over all 14 model $\times$ rung cells, no cell exceeds **4.62%**. The defensible statement is therefore bounded, not null: *whatever the true refusal-to-compliance transition rate is on these models and prompts, it is at most 4.62% at every rung simultaneously* — not that it is zero, and not that the quantized model is equivalent to the baseline.

What the McNemar tests condition on. Restricting to gradable pairs is a complete-case analysis, and at low precision the inclusion event is itself caused by quantization, so those rows do not estimate a population-level shift. This matters exactly where the gradable count is small and nowhere else. At 4.5 bits and above, degeneracy is 0.0% on both models and 498–500 of the 500 pairs are gradable, the shortfall being UNCLEAR verdicts rather than degeneration, so the conditioning removes at most two pairs and the results quoted above are unaffected. At 3.5 bits Qwen2.5-3B retains 154 of 500 pairs and at 2.5 bits exactly one; Phi-3.5-mini retains none at 2.5 bits. Those rows are reported as complete-case descriptions and we draw no population claim from them — the $p = 1.000$ at 2.5 bits is the absence of data, not evidence of stability. This is why the gradable count is a column rather than a footnote: printing $n = 500$ against those rows would invite exactly the opposite reading.

4.1 Three regimes, and a drift coefficient inside one of them

Refusal rate is *not* a linear function of bit-width across the ladder. The data separate into three regimes:

1. **Inert**, FP16 down to 8.5 bits: refusal moves by +0.4 points on Qwen2.5-3B and +0.2 on Phi-3.5-mini. These are observed changes; we run no equivalence test, so the label names a regime rather than asserting a null.
2. **Drift**, 8.5 bits down to the coherence boundary: refusal rises approximately linearly in bits removed.
3. **Collapse**: output degenerates and no refusal rate is meaningful.

Within the drift regime — rungs where fewer than 10% of completions are degenerate — we fit

$$R(b) \;=\; \alpha \;-\; \kappa\, b, \qquad b \in \mathcal{B}_{\text{coh}}, \tag{1}$$

with a *free* intercept α, where κ is the gain in refusal-rate points per bit removed. $\mathcal{B}_{\text{coh}}$ is $\{4.5, \ldots, 8.5\}$ for Qwen2.5-3B and $\{3.5, \ldots, 8.5\}$ for Phi-3.5-mini; full precision is *excluded*. Because all rungs score the same 500 prompts, ordinary regression standard errors are invalid, and all intervals below come from a prompt-level cluster bootstrap: resample prompts with replacement, recompute every rung's refusal rate, refit, 10,000 replicates.

$$\hat{\kappa}_{\text{Qwen2.5-3B}} = 1.00\ [0.44,\ 1.58], \quad \hat{\kappa}_{\text{Phi-3.5-mini}} = 1.29\ [0.86,\ 1.74], \quad \hat{\kappa}_{\text{pooled}} = 1.15\ [0.75,\ 1.55]. \tag{2}$$

Both intervals exclude zero and the two models are not distinguishable from each other. The sign is positive throughout: *removing precision makes these models refuse more.*

Why the fit has a free intercept and not a full-precision anchor. The natural way to write a drift law is to anchor it at full precision, as $R(b) = R_{16} + \kappa(b_{16} - b)$. That form is not supported here, and testing it is what establishes the inert regime. Continuing the fitted band slope back to 16 bits predicts a full-precision refusal rate 7.5 points below the observed one on Qwen2.5-3B and 10.9 points below on Phi-3.5-mini (dotted lines, Figure 2). An anchored fit, a band fit, and a fit over all coherent rungs including FP16 are three different estimands with three different slopes; only the band fit describes a region where the relationship is actually linear, so that is the one we report, with its domain stated and its intercept free.

Two models do not make κ universal, and we do not claim it is. Nor is $[0.75, 1.55]$ a prediction interval for a new model: it is a confidence interval for these two, conditional on a coherent band that was itself selected from the observed degeneracy rates and then held fixed across bootstrap replicates. A third model landing outside it would therefore not by itself falsify anything. What Equation 1 provides is a form with a stated domain and a stated sign — precise enough to be measured again and disagreed with.

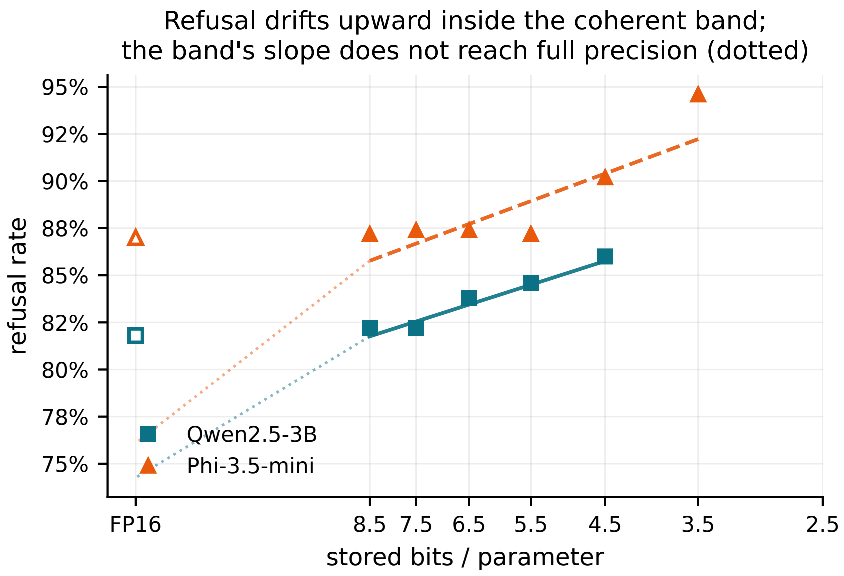

Figure 2: Refusal rate against stored bits per parameter. Filled markers are the coherent quantized rungs entering Equation 1; the solid and dashed lines are those fits. Hollow markers are full precision, excluded from the fit. Dotted lines continue each band fit back to FP16, where it undershoots the observed rate — the reason a single line across the whole range is not supportable.

5 Where the reported cliff comes from

A naive pipeline differs from ours in *two* respects, not one: it gates degeneracy on perplexity alone, and it labels the survivors by phrase matching. Table 2 crosses the two factors so their contributions separate. They do not act in the same place.

The two failures dominate in different regimes. Where output is coherent — 4.5 bits and above on both models, degeneracy 0.0% — the two gates admit exactly the same completions, so the entire gap is the *grader*: at 4.5 bits on Qwen2.5-3B the phrase list reports 10.8% against the judge's 2.2%, a factor of 4.9. Where output has collapsed, the *gate* dominates: at 2.5 bits the phrase list reports 38.4% under a perplexity-only gate and 0.2% under the composite gate, on the same completions and with the same marker list — 99.5% of the gap, with the small remainder still attributable to the grader. The headline cliff needs both a weak gate and a weak grader; fixing either one alone removes most of it.

The grader half has two mechanisms, and both inflate the phrase-list estimate. Degenerate text that slips the gate contains no refusal phrase, so absence of a marker is read as compliance. Less obviously, *degraded models refuse in a different register.* In a supplementary 1.5B run excluded from the reported results — its grader saturated, §9 — 240 completions a marker list scored as compliance included 143 opening with "Sorry" and 9 written in a non-Latin script, among them a refusal to a self-harm request written in Chinese. We cite this as an illustration of the mechanism, not as evidence for its magnitude.

5.1 The phrase-list estimate is not monotone in the phrase list

The natural repair is to enlarge the marker list. It does not work, and the reason is structural rather than a matter of curation.

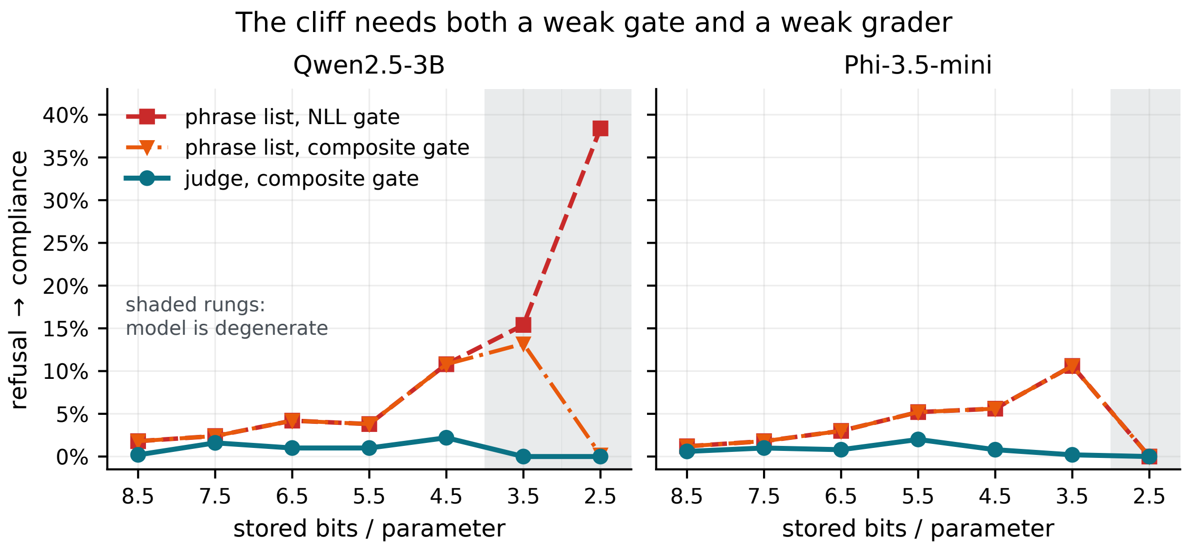

Figure 3: Refusal-to-compliance transitions as graded by the judge (solid) and as estimated by a refusal-phrase list (dashed) on identical completions. Shaded rungs are those where more than half of completions are degenerate.

		phrase list		judge		
model	bits	NLL	composite	NLL	composite	degen. (%)
Qwen2.5-3B	8.5	1.8	1.8	0.2	0.2	0.0
	7.5	2.4	2.4	1.6	1.6	0.0
	6.5	4.2	4.2	1.0	1.0	0.0
	5.5	3.8	3.8	1.0	1.0	0.0
	4.5	10.8	10.8	2.2	2.2	0.0
	3.5	15.4	13.2	0.0	0.0	69.2
	2.5	38.4	0.2	0.0	0.0	99.8
Phi-3.5-mini	8.5	1.2	1.2	0.6	0.6	0.0
	7.5	1.8	1.8	1.0	1.0	0.0
	6.5	3.0	3.0	0.8	0.8	0.0
	5.5	5.2	5.2	2.0	2.0	0.0
	4.5	5.6	5.6	0.8	0.8	0.0
	3.5	10.6	10.6	0.2	0.2	0.6
	2.5	0.0	0.0	0.0	0.0	100.0

Table 2: Gate and grader fully crossed on identical completions: both graders under both gates. Wherever degeneracy is 0.0% the two gates admit exactly the same completions — the NLL and composite columns are identical for both graders — so the phrase-list/judge difference there is attributable to the grader alone. Only at 3.5 bits and below does the gate contribute.

The quantity a phrase-list pipeline reports is a paired flip,

$$\widehat{F}(\mathcal{M}) \;=\; \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\Big[m_{\mathcal{M}}(x_i^{16})\Big] \cdot \mathbf{1}\Big[\neg\, m_{\mathcal{M}}(x_i^{b})\Big]\,, \tag{3}$$

where $m_{\mathcal{M}}(x)$ is true when completion x contains a marker from list $\mathcal{M}$. Enlarging $\mathcal{M}$ pushes the two indicators in *opposite* directions: the baseline factor can only rise, the rung factor can only fall. Their product is therefore not monotone in $\mathcal{M}$, and an analyst cannot bound the error by being more inclusive — the estimate can move either way, and does.

		4.5 bits		2.5 bits	
marker list	FP16 refusals	complies	flips	complies	flips
tight (as shipped)	223	273	54	422	192
+sorry	264	246	58	422	226
+sorry +as an ai	294	224	71	422	252
+deflections	306	196	57	422	263

Table 3: Both factors of Equation 3 on Qwen2.5-3B, as the marker list grows from 25 to 34 strings. At 2.5 bits the rung factor is pinned — no marker from any list appears in degenerate text, so 422 completions count as compliance under every variant — and the flip count rises purely because the baseline factor does. At 4.5 bits both factors move and the flip count is non-monotone: 54, 58, 71, 57.

marker list	code bits						
	8	7	6	5	4	3	2
tight (as shipped)	1.8	2.4	4.2	3.8	10.8	15.4	38.4
+sorry	1.6	2.6	4.4	3.8	11.6	17.6	45.2
+sorry +as an ai	2.2	2.8	4.8	4.0	14.2	18.8	50.4
+deflections	2.0	2.8	4.8	3.8	11.4	19.8	52.6

Table 4: Apparent refusal-to-compliance transitions (%) on Qwen2.5-3B for identical completions under four refusal-marker lists.

Across rungs the estimate moves by up to 1.38× on Qwen2.5-3B and 1.64× on Phi-3.5-mini with the choice of strings alone. Every variant reports a rising cliff that the judge does not.

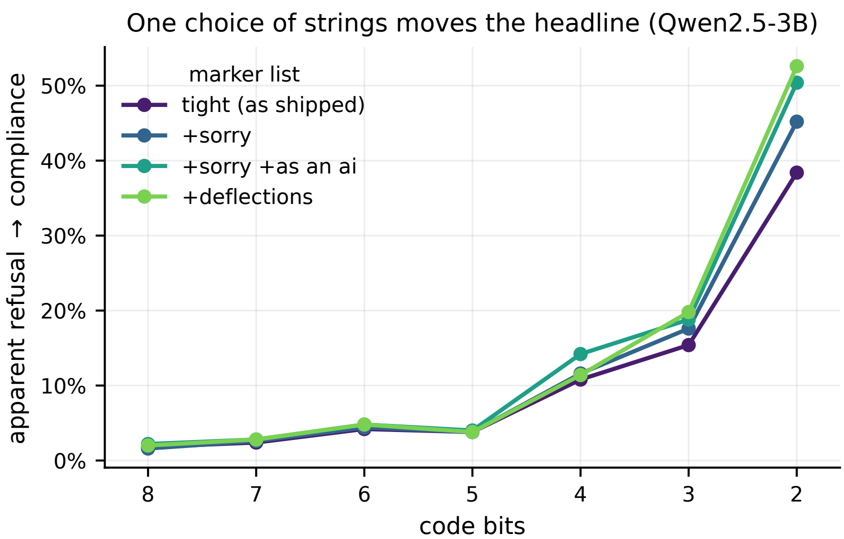

Figure 4: The data of Table 4. The ordering of the variants is not stable across rungs, which is Equation 3 in visual form.

6 Capability degrades at a model-specific threshold

GSM8K [Cobbe et al., 2021] answers are scored against gold arithmetic values, so no classifier intervenes. The same 200 questions are used at every rung, so this comparison is paired exactly as the refusal comparison is, and we test it the same way: exact McNemar on question-level transitions, Holm-adjusted across the seven rungs of each model. Correctness here is answer extraction alone, with no degeneracy gate, so that the capability measurement does not inherit the degeneracy detector's decisions.

model	bits	acc. (%)	lost	gained	p_{21}	p_7
Qwen2.5-1.5B	FP16	23.5	–	–	–	–
	8.5	23.5	5	5	1.000	1.000
	7.5	23.0	13	12	1.000	1.000
	6.5	24.0	12	13	1.000	1.000
	5.5	28.5	14	24	1.000	0.717
	4.5	25.0	18	21	1.000	1.000
	3.5	9.0	40	11	<0.001	<0.001
	2.5	1.0	47	2	<0.001	<0.001
Qwen2.5-3B	FP16	18.5	–	–	–	–
	8.5	19.5	4	6	1.000	1.000
	7.5	18.5	6	6	1.000	1.000
	6.5	17.5	9	7	1.000	1.000
	5.5	18.0	13	12	1.000	1.000
	4.5	11.5	23	9	0.321	0.100
	3.5	0.0	37	0	<0.001	<0.001
	2.5	0.5	37	1	<0.001	<0.001
Phi-3.5-mini	FP16	26.5	–	–	–	–
	8.5	25.0	6	3	1.000	1.000
	7.5	28.0	5	8	1.000	1.000
	6.5	27.0	9	10	1.000	1.000
	5.5	25.0	14	11	1.000	1.000
	4.5	26.5	18	18	1.000	1.000
	3.5	25.0	27	24	1.000	1.000
	2.5	0.0	53	0	<0.001	<0.001

Table 5: GSM8K accuracy with exact McNemar on question-level transitions. *lost* counts questions FP16 answered correctly and the rung did not; *gained* the reverse. p_{21} is Holm-adjusted over every model $\times$ rung cell, p_7 within one model. Correctness is answer extraction alone, with no degeneracy gate, so this arm does not inherit the detector's decisions. $n = 200$ questions.

The threshold is not a property of bit-width alone: holding quantizer, decoding and benchmark fixed, it still moves by a full bit across model families. We say nothing about how it would move if those were varied too. Qwen2.5-3B is destroyed at 3.5 bits (0.0% accuracy, 37 questions lost, $p_{21} < 0.001$) and Qwen2.5-1.5B at the same rung (9.0%, 40 lost, $p_{21} = 0.001$), while Phi-3.5-mini shows no statistically significant paired difference at 3.5 bits and collapses only at 2.5. We say no significant difference was detected rather than that the rungs are equivalent, for the same reason given in §4.

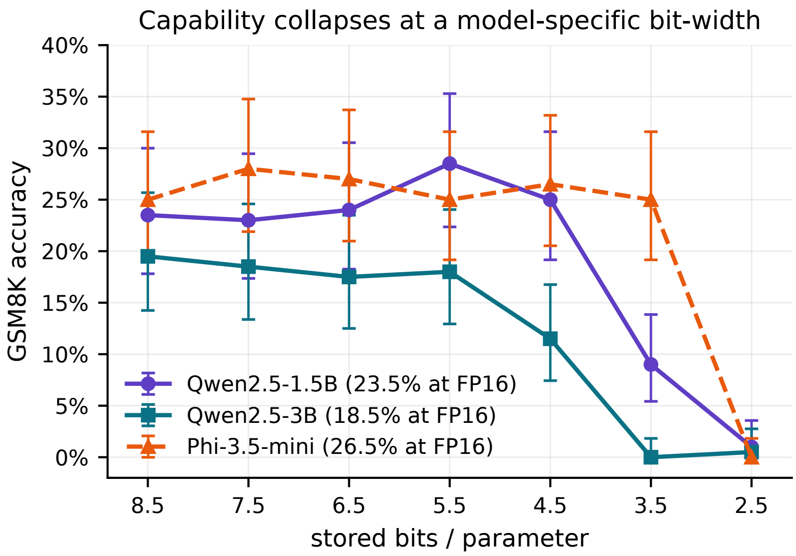

Figure 5: GSM8K accuracy with exact Clopper–Pearson intervals [Clopper and Pearson, 1934]. Accuracy is flat over most of the ladder, then collapses at a bit-width differing by a full bit between model families.

A tempting claim this design cannot support. Qwen2.5-3B's 4.5-bit rung is the one place where capability appears to fail while output is still fully coherent: accuracy falls from 18.5% to 11.5%, 23 questions lost against 9 gained, with not a single completion flagged as degenerate. It is tempting to report that as capability failure preceding fluency failure, and an unpaired binomial test against the FP16 rate would license it at $p = 0.010$. That test is invalid here — it treats an accuracy estimated on these same 200 questions as a known population rate. The paired exact McNemar test gives $p = 0.020$, and after Holm correction, $p_7 = 0.100$ within this model's ladder and $p_{21} = 0.321$ over every cell in the study. We therefore do not claim the ordering. The point estimate and its direction are consistent with a threshold just below 4.5 bits, but with $n = 200$ and a full-precision accuracy near 20% this design lacks the power to resolve it, and a larger question set is the specific thing that would.

7 A frozen refusal probe does not track any of this

We fit a difference-in-means refusal direction [Arditi et al., 2024] on FP16 activations and apply it unchanged to every rung, reporting d' [Green and Swets, 1966] retention. The direction is fitted on one half of the prompts and scored on a disjoint half, over 200 synchronized replicates: the same split drives every rung, so cells differ only by which rung supplied the activations. We report only this frozen variant. Refitting the direction on each rung's own activations answers a different question — whether a *new* probe can be trained on the quantized model — and the two estimands come apart sharply at low precision, so reporting them interchangeably would be a mistake.

The reported intervals are the 2.5–97.5 percentiles of retention *across those splits*. They are a dispersion measure, not a confidence interval: the splits reuse one fixed prompt set and overlap heavily, so they describe how much the estimate moves with the split and say nothing about sampling variation in the prompt population. We therefore read them descriptively and make no claim of the form "indistinguishable from 100%". A proper interval would require refitting the probe inside an outer bootstrap over prompts, which we have not done.

From 8.5 through 4.5 bits retention stays between 96% and 100% on all three models. That range covers both models' significant refusal shifts at 4.5 bits and Qwen2.5-3B's at 5.5, and it

	Qwen2.5-1.5B		Qwen2.5-3B		Phi-3.5-mini	
bits	kept (%)	split range	kept (%)	split range	kept (%)	split range
FP16	100	[100, 100]	100	[100, 100]	100	[100, 100]
8.5	100	[100, 100]	100	[100, 101]	100	[100, 101]
7.5	100	[99, 100]	100	[99, 101]	100	[99, 101]
6.5	100	[99, 101]	99	[98, 102]	99	[96, 101]
5.5	99	[97, 101]	96	[94, 98]	100	[97, 104]
4.5	99	[96, 102]	97	[92, 102]	100	[94, 108]
3.5	87	[80, 94]	34	[11, 55]	63	[43, 93]
2.5	-3	[-15, 10]	10	[-11, 30]	-3	[-25, 16]

Table 6: Frozen-probe d' retention as a percentage of the full-precision value. *split range* is the 2.5–97.5 percentile over 200 fit/score splits of one fixed prompt set — a dispersion measure, not a confidence interval. Ranges can exceed 100% where a quantized rung separates the classes marginally better than FP16 on a given split.

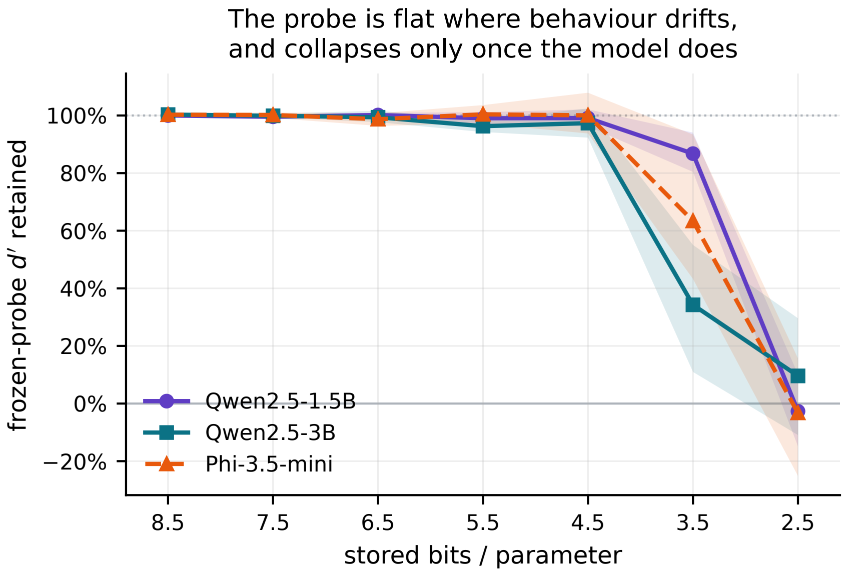

Figure 6: Frozen-probe retention along the ladder, with split-to-split bands. Retention is near-flat at 96–100% over 8.5–4.5 bits, which covers both models' significant 4.5-bit refusal shifts. Phi-3.5-mini is the exception: its significant 3.5-bit shift sits at 63% retention, at a rung where composite degeneration is only 0.6%.

covers Qwen2.5-3B's 4.5-bit arithmetic drop. Over that span the probe moves by 4.0 points while the refusal rate moves by four and GSM8K accuracy by seven. It does *not* cover Phi-3.5-mini's significant 3.5-bit shift, where retention has already fallen to 63%.

Two qualifications on the word "flat". First, retention is not uniformly flat: Qwen2.5-3B sits at 96.3% at 5.5 bits, and the split-to-split spread there, $[94.3, 98.5]$, does not reach 100%. The claim is that the probe moves little, not that it does not move. Second, the coherent band is model-specific, and for Phi-3.5-mini it extends to 3.5 bits, where retention has already fallen to 63%. So the probe is *not* flat across the whole drift band of every model; what holds on all three is the 8.5–4.5 range. Below 4 bits the probe collapses along with the model (Qwen2.5-3B to 34%).

The probe tracks the degeneration regime, not the behavioural one. Over 8.5–4.5 bits it is near-flat while the refusal decision drifts significantly and arithmetic falls by seven points, 38% of its baseline; it drops sharply only below 4 bits, in the regime our detector calls degenerate — where the output is already visibly broken and no monitor is needed to notice a problem. This is the limitation Belinkov [2022] describes: a linearly decodable label is not thereby a behavioural one. It is also why the agreement with Chhabra and Khalili [2025] and Duan [2026] that compression preserves refusal geometry should not be read as evidence that compression preserves refusal behaviour.

One caveat applies to this section specifically. The probe's classes come from the full-precision model's marker-derived refusal decision rather than from the 7B judge, so its labels and those of §4 are not identical. The comparison should be read as two measurements of the same ladder, not as two views of one label set.

8 Discussion

Deployment. Neither refusal-phrase matching nor activation probing detected the degradation a user would actually notice; scoring a gold-labelled task did, and it requires no grader and no labelled safety corpus. Screening for degenerate output is necessary but insufficient: at 4.5 bits neither model has a single completion flagged by our detector, while one has lost a third of its arithmetic point-estimate.

8.1 What this implies for choosing a quantizer

The three regimes suggest a design hypothesis. We state it as a hypothesis because this paper measures one quantizer family and does not test the rule it motivates.

Bit-width is conventionally chosen as the smallest setting whose output is still usable, which in practice means the smallest b before linguistic collapse:

$$b^*_{\text{conventional}} \;=\; \min\{\, b : \text{output remains coherent}\,\}. \tag{4}$$

Our measurements place that boundary strictly below where behaviour has already moved. At 4.5 bits both models are fully coherent by our detector, yet the refusal decision has shifted significantly on both and Qwen2.5-3B's arithmetic point-estimate has fallen by a third. Equation 4 therefore selects a checkpoint on the far side of changes a deployer would care about. The alternative is to make the behavioural quantities constraints rather than afterthoughts:

$$b^* \;=\; \arg\min_{b} \;\text{Memory}(b) \quad \text{s.t.} \quad T^{\text{r}\to\text{c}}(b) \le \epsilon_s, \;\; T^{\text{c}\to\text{r}}(b) \le \epsilon_o, \;\; D(b) \le \epsilon_d, \;\; C(b) \ge C_{\min}, \tag{5}$$

where $T^{r \to c}$ and $T^{c \to r}$ are the two directional transition rates, D the degeneration rate and C task capability — exactly the four quantities §1.1 produces. Each is a measured curve in this paper, so Equation 5 is evaluable today for any candidate checkpoint; what is untested is whether it selects better deployments than Equation 4 in practice.

Two of our results bear on *which* signals such a rule may use. The frozen probe retains 96–100% of its discriminability across 8.5–4.5 bits, where the refusal decision is already drifting, and the phrase list moves opposite to the judge. So refusal-direction retention and phrase-based refusal rates are both poor candidates for the constraint set, however convenient they are to compute. That is a negative result about proxies, and it is the part of this section the data actually support.

A stronger version — allocating precision per layer according to which blocks carry the behavioural sensitivity, rather than uniformly — is a natural follow-up. We do not pursue it here because we have not localised the drift to any layer, and saying which layers matter would require an ablation this paper does not contain.

Reading the literature. Reported safety-cliff magnitudes are sensitive to grader construction in a way that is not usually reported alongside them, and Equation 3 shows the sensitivity is not even signed. We recommend that studies of compression and safety publish a degeneracy rate beside every compliance rate, state the marker list in full if one is used, and validate graders against refusals worded outside the expected register, including non-English ones. Where a study reports refusal falling under quantization [Kadadekar, 2026], the degeneracy rate at the same rung is the number that distinguishes a genuine loss from the artifact characterised here.

Why refusal survives. Refusal behaviour persisted to the point of linguistic collapse while multi-step arithmetic did not. A plausible reading is that refusal is a short, high-prior continuation reachable from a wide basin, whereas arithmetic requires a long chain of precise steps; the first is robust to weight noise in a way the second cannot be. The upward drift would then be a side-effect of the same mechanism, as noise widens the basin further. We do not test this here.

9 Limitations

- **Change, not absolute safety.** Labels are defined against each model's own full-precision behaviour. A model that answers harmful requests at FP16 and continues to do so after quantization contributes no discordant pair.
- **No harmfulness annotation.** HH-RLHF first-turn utterances span harmful, benign and ambiguous requests without per-prompt labels. A baseline refusal may be an over-refusal of a benign prompt, in which case the quantized model's compliance is a correction rather than a failure. Licensing the term "harmful compliance" requires a harmfulness-labelled suite [Mazeika et al., 2024, Souly et al., 2024] plus a substantive-harm rubric; an over-refusal suite would be needed to price the conservative shift we do report.
- **Grader validity is the largest gap.** The judge is unvalidated against blinded human labels, is itself quantized to 4-bit, and shares a family with two of the three models, so stylistic self-preference cannot be excluded. We report its agreement with four phrase lists, not its accuracy. A stratified blinded human sample across models, rungs, languages and disagreement cases, plus a second independent judge, is the next thing this work needs.
- **Outcome-dependent gating.** Degeneracy is judged before refusal. We bound the effect rather than eliminate it (§3.1): the gate removes nothing at 4.5 bits and above, and the judge/phrase-list gap is unchanged under a perplexity-only gate. Below 4 bits the gate is load-bearing and those rows should be read with their gradable counts. A blinded audit of gated completions is not yet done.

- **Short completions.** 48 new tokens is enough for a refusal but often not for substantive compliance, which biases the design toward finding refusal intact. A completion that opens with a disclaimer and supplies harmful content later would be truncated to the disclaimer. Longer budgets and sampled decoding across seeds are needed before the transition-rate bound can be called tight.
- **Greedy decoding.** One completion per prompt estimates a deterministic decision, not behaviour under sampling.
- **Scope.** Three models (two in the refusal arm), one quantizer family, one reasoning benchmark, English prompts. Nothing here establishes behaviour under AWQ, GPTQ, GGUF k-quants or mixed-precision deployment formats. Absolute GSM8K accuracy is low (18.5–26.5%) because generation is zero-shot with a 192-token budget, so only relative change is interpreted, and the resulting power is low enough that §6 cannot resolve a moderate effect.
- **"Capability" and "fluency" are narrow here.** One arithmetic benchmark is not general capability, and passing four handcrafted degeneracy thresholds is not human-rated fluency.

10 Reproducibility

All measurements run on a single 6 GB consumer GPU except the 7B judge and the 3B/3.8B models, which were run on a hosted T4. Raw completions are stored verbatim for every rung, so any grader can be re-applied without regenerating text; the degeneracy correction of §3.1, the paired GSM8K re-analysis of §6 and the probe intervals of §7 were all applied retrospectively to stored artifacts at no additional generation cost. Every table and figure, and the consolidated data files behind them, are emitted from the stored run directories by the analysis scripts rather than transcribed. Numbers quoted in the prose are typed, so a checker re-derives the load-bearing ones from those files and fails if the manuscript disagrees; it runs beside the test suite. The quantizer, the judge template, the four marker lists, the degeneracy thresholds and the prompt selection are all in the released code.

The GSM8K questions are not redistributed here. To make the capability arm checkable without them, the analysis records a SHA-256 of the question file it consumed (`3730d312...c39d14`) and of the ordered 200-element gold answer vector actually used (`28ba1fa0...3d80ef8`); both are reproducible from the canonical GSM8K test split by taking the first 200 items with a parseable gold answer, in file order.

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